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GENERATIVE AI & PYTHON BASICS

A hands-on program for **engineering students, college students and working professionals**, technical and non-technical. GenAI is taught with **no-code tools**; Python is taught as **literacy**, so learners can read, run and check the code AI writes for them.

EVERY SESSION = 3 HOURS

SAME PATTERN ACROSS ALL PROGRAMS

2 hrs
Concept & live demo
Trainer-led, with real examples

1 hr
Practical
Learners build it themselves

Each session ends with something the learner has built: a prompt set, a chatbot, a workflow, an agent or a working notebook.

PROGRAM OUTLINE

FOUR FORMATS · GENAI + PYTHON BASICS

Program	Mode	Schedule	Duration	Sessions · Hours	Projects
Workshop	Online / Offline	1 day, 1 session	3 hours	1 · 3 hrs	Hands-on build
Winter Crash Course	Online / Offline	2–3 days a week	10 weeks about 2.5 months	20–30 · 60–90 hrs	Mini project
3-Month Course	Online / Offline	2–3 days a week	12 weeks	24–36 · 72–108 hrs	3 monthly capstones
6-Month Course	Online	2–3 days a week	24 weeks	48–72 · 144–216 hrs	6 monthly capstones

CURRICULUM MODULES

WHAT EACH PROGRAM DRAWS FROM

Generative AI

No-code tools throughout

- GenAI Foundations & the LLM Landscape
- Prompt Engineering: Basics to Advanced Patterns
- AI Reliability, Safety & Evaluation
- AI Productivity & Meeting Intelligence
- Web Research with Citations
- Knowledge Management & RAG
- No-Code AI Chatbots
- AI for Content Creation
- Images, Video & Voice with AI
- Workflow Automation with n8n & Zapier
- AI Agents: Tools, Memory & MCP
- Building Agents Without Code
- Vibe Coding with No-Code IDEs
- Claude Skills, Artifacts & claude.md
- Prompt Management as a Practice
- AI System Design & Going Live

Python Basics

In Google Colab, with an AI assistant

- Data Types, Variables & Input/Output
- Operators
- Control Statements
- Loops
- Functions
- Lists
- Strings
- Sets, Tuples & Dictionaries
- File Handling
- Regular Expressions
- Functional Programming
- Object-Oriented Programming

PROGRAM TIMELINE

CONCEPT (2 HRS) AND PRACTICAL (1 HR) FOR EVERY SESSION · A RESUME-READY CAPSTONE AT THE END OF EVERY MONTH

WORKSHOP

1 day | 3 hours | Online / Offline

Time	Session	Learners walk away with
Hour 1	GenAI foundations: how LLMs work, the tool landscape, safe use Concept & demo	A clear map of which AI tool to use when
Hour 2	Prompt engineering: roles, context, examples, step-by-step reasoning Concept & demo	A prompt framework they can reuse
Hour 3	Practical: build a personal prompt library and a no-code chatbot on their own document Hands-on	5 tested prompts + a working chatbot

WINTER CRASH COURSE

10 weeks | **2–3** sessions a week | **3** hrs a session | Online / Offline

Week	Concept & demo (2 hrs)	Practical (1 hr)
Week 1	GenAI foundations & the LLM landscape	Compare three AI tools on one real task
Week 2	Prompt engineering: basics to advanced patterns	Build a personal prompt library
Week 3	AI reliability, safety & web research with citations	Fact-check AI output; write a cited brief
Week 4	Python I: data types, variables, I/O, operators	Set up Colab; run and edit an AI-written script
Week 5	Python II: control statements, loops, functions	Batch-process prompts with a loop
Week 6	Knowledge management, RAG & no-code chatbots	Chatbot that answers from your own documents
Week 7	AI for content creation: images, video & voice	Poster, voice-over and short video
Week 8	Workflow automation with n8n & Zapier	Form → AI summary → email workflow
Week 9	AI agents: concepts, tools & no-code build	Build a first research or study agent
Week 10	Vibe coding a simple app + project review	Mini project demo day

3-MONTH COURSE

12 weeks | **2–3** sessions a week | **3** hrs a session | **3** capstones | Online / Offline

Week	Concept & demo (2 hrs)	Practical (1 hr)
MONTH 1 · GENAI FOUNDATIONS & PYTHON START		
Week 1	GenAI foundations, LLM landscape & AI productivity tools	Set up a personal AI toolkit
Week 2	Prompt engineering: anatomy to advanced patterns	Prompt library for study or work
Week 3	AI reliability, safety, research & citations	Cited research report
Week 4	Python: data types, variables, I/O, operators, control statements	Read and modify AI-written scripts in Colab
Capstone 1	**AI Research & Productivity Assistant** A tested prompt library plus a Colab script that researches a topic and writes a cited summary	**RESUME LINE** Built an AI research assistant with a reusable prompt library and Python automation
MONTH 2 · PYTHON, KNOWLEDGE & CONTENT		
Week 5	Python: loops, functions, lists & strings	Reusable prompt function + batch run
Week 6	Python: sets, tuples, dictionaries & file handling	Summarise a CSV or PDF with AI
Week 7	Knowledge management, RAG & no-code chatbots	Deployed FAQ chatbot
Week 8	AI for content creation: images, video & voice	Campaign kit: post, image, voice, video
Capstone 2	**Document Q&A Chatbot** A live chatbot that answers from a real college, shop or company document, with a launch poster and demo video	**RESUME LINE** Deployed a RAG-based chatbot that answers questions from custom documents
MONTH 3 · AUTOMATION, AGENTS & FINAL CAPSTONE		
Week 9	Workflow automation with n8n & Zapier	Lead or feedback automation
Week 10	AI agents: tools, memory & MCP	Agent design canvas
Week 11	Building agents without code + vibe coding	Working agent or mini app
Week 12	Prompt management & capstone review	Capstone build sprint
Capstone 3	**AI Agent for a Real Workflow** An agent plus automation that solves one real task end to end, presented on demo day	**RESUME LINE** Built and demoed a no-code AI agent that automates a real business workflow

6-MONTH COURSE

24 weeks | **2–3** sessions a week | **3** hrs a session | **6** capstones | Online

Week	Concept & demo (2 hrs)	Practical (1 hr)
MONTH 1 · GENAI FOUNDATIONS & PROMPTING		
Week 1	GenAI foundations & the LLM landscape	Compare AI tools on real tasks
Week 2	AI productivity & meeting intelligence	Meeting-notes and email workflow
Week 3	Prompt engineering: anatomy & core patterns	Prompt library, version 1
Week 4	Advanced prompt patterns & prompt management	Tested, versioned prompt library
Capstone 1	**Personal AI Productivity Kit** A versioned prompt library plus a meeting-notes-to-action-items workflow	**RESUME LINE** Designed a tested prompt library and AI productivity workflow
MONTH 2 · PYTHON BASICS I		
Week 5	Data types, variables & input/output	Colab setup; first AI-written script
Week 6	Operators & control statements	Trace and fix decision logic
Week 7	Loops & functions	Batch-prompt notebook
Week 8	Lists & strings	Clean text before sending it to AI
Capstone 2	**AI Batch Summariser Notebook** A Colab notebook that loops through many inputs, cleans the text and summarises each with AI	**RESUME LINE** Built a Python notebook that batch-processes text with an LLM
MONTH 3 · PYTHON BASICS II & AI RELIABILITY		
Week 9	Sets, tuples & dictionaries	Read JSON returned by AI tools
Week 10	File handling & regular expressions	Extract data from files and AI output
Week 11	Functional programming & OOP: reading the code	Explain an AI-written program line by line
Week 12	AI reliability, safety & evaluation	Evaluation checklist for an AI answer
Capstone 3	**Smart Document Extractor** Reads CSV or PDF files, pulls out key fields with AI and regex, and checks the output for errors	**RESUME LINE** Built an AI data-extraction tool with output validation in Python
MONTH 4 · KNOWLEDGE, CHATBOTS & CONTENT		
Week 13	Web research with citations	Cited research report
Week 14	Knowledge management & RAG	Document Q&A assistant
Week 15	No-code AI chatbots	Deployed FAQ chatbot
Week 16	Content creation: images, video & voice	Content campaign kit
Capstone 4	**Knowledge Chatbot with Launch Kit** A deployed chatbot on real documents, with a poster, voice-over and demo video to launch it	**RESUME LINE** Deployed a RAG chatbot and produced its AI-generated launch content
MONTH 5 · AUTOMATION & AI AGENTS		
Week 17	Workflow automation essentials	Map a real workflow end to end
Week 18	Automating with n8n & Zapier	Working automation with AI steps
Week 19	AI agents: tools, memory & MCP	Connect an agent to tools

Week	Concept & demo (2 hrs)	Practical (1 hr)
Week 20	Building agents without code & multi-agent systems	Personal assistant agent
Capstone 5	**Business Automation Agent** An n8n or Zapier workflow with an AI agent that handles a real process, such as leads, bookings or support	RESUME LINE Automated a business process with an AI agent and n8n workflows
MONTH 6 · BUILD, DEPLOY & CAPSTONE		
Week 21	Vibe coding with no-code IDEs	Mini web app
Week 22	Claude skills, artifacts & claude.md	Reusable skill for a real task
Week 23	AI system design & going live	Host the capstone online
Week 24	Capstone review & portfolio	Capstone build sprint
Capstone 6	**Live AI Application** A hosted AI app solving a real problem, presented on demo day with a portfolio of all six capstones	RESUME LINE Designed, built and deployed a live AI application end to end

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